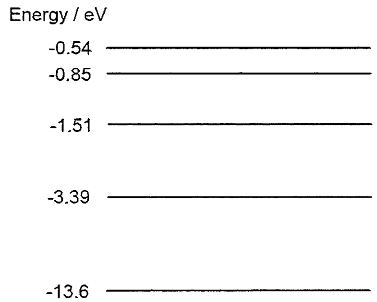
	Raffles Institution Physics Department Revision Package 1	Quantum Physics	19

Important note: ***denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs

1. **[AJC/H2/Prelims 2010/P1/35]**

The figure shows the energy levels of the hydrogen atom.



What is the maximum number of emission spectral lines observed if electrons of energy 12.9 eV are incident to the hydrogen atom at the ground state?

A 0 B 3 C 6 D 10

2. **[CJC/H2/Prelims 2010/P1/34]**

What is the de Broglie wavelength of a rifle bullet of mass 0.02 kg which is moving at a speed of 300 m s^{-1}?

A 7.3×10^{-36} m B 1.8×10^{-35} m C 1.1×10^{-34} m D 9.9×10^{-33} m

3. **[CJC/H2/Prelims 2010/P1/35]**

Which statement is true?

A A beam of electrons directed at a vessel of cold gas can cause the formation of either an absorption or emission line spectrum.

B A beam of white light directed at a vessel of cold gas can cause the formation of either an absorption or emission line spectrum.

C A beam of electrons directed at a vessel of cold gas can only cause the formation of an absorption line spectrum.

D A beam of electrons directed at a vessel of cold gas can only cause the formation of an emission line spectrum.

4. **[Dunman High/H2/Prelims 2010/P1/36]**

Electrons emitted by a hot filament pass down a tube containing hydrogen and are then collected by an anode which is maintained at a positive potential with respect to the filament. The gas near the anode is found to emit monochromatic ultra-violet radiation. Given that the energy levels of an electron in a hydrogen atom is

$$E/\text{eV} = \frac{13.6}{n^2}, \text{ where } n = 1, 2, 3, \ldots$$

What is the wavelength of the monochromatic ultra-violet radiation?

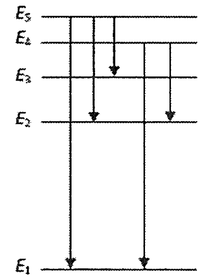
A 122 nm B 102 nm C 97 nm D 91 nm

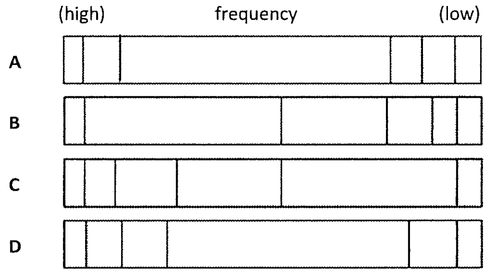
5. **[Dunman High/H2/Prelims 2010/P1/35]**

The figure on the right shows five energy levels of an atom, one being much lower than the other four.

Five transitions between the levels are indicated, each of which produces a photon of definite energy and frequency.

Which one of the spectra below best corresponds to the set of transitions indicated?





6. **[NJC/H2/Prelims 2010/P1/36]**

Which statement is true about excitation of electron in an atom?

A An electron will be excited as long as the incoming free electron kinetic energy is greater than the energy difference between the initial and any other energy states of the atom.

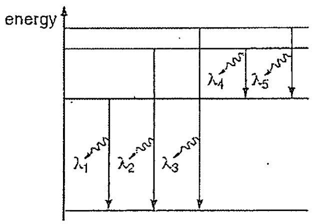
B Electron will be excited as long as the photon's energy is of the same value as any of the energy state of the atom.

C When visible white light is shone onto an insulator, electrons are excited.

D Excitation of electrons will only take place when two hydrogen atoms collide elastically.

7. **[IJC/H2/Prelims 2010/P1/35]**

The diagram shows the energy levels for an atom, drawn to scale. The electron transitions give rise to the emission of a spectrum of lines of λ_1, λ_2, λ_3, λ_4 and λ_5.



What can be deduced from this diagram?

A $\lambda_1 > \lambda_2$

B $\lambda_3 = \lambda_4 + \lambda_5$

C λ_4 is the shortest of the five wavelengths.

D The transition corresponding to wavelength λ_3 represents the ionisation of the atom.

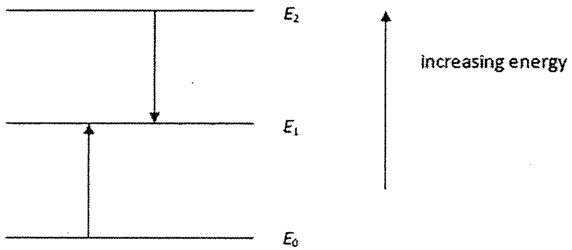
8. **[MI/H2/Prelims 2010/P1/35]**

A laser beam of power P has wavelength λ. What is the rate of photons produced?

A $\dfrac{hc}{P\lambda}$ B $\dfrac{Ph\lambda}{c}$ C λPhc D $\dfrac{P\lambda}{hc}$

9. **[NYJC/H2/Prelims 2010/P1/36]**

The diagram shows part of the energy level picture of a particular element. The energy change for E_0 to E_1 is the same as that for E_1 to E_2.



If the transition E_2 to E_1 corresponds to a red line in the element's spectrum, then the transition E_o to E_1 corresponds to

A absorption of red light.

B emission of red light.

C absorption of infra-red radiation.

D emission of infra-red radiation.

10. **[RVHS/H2/Prelims 2010/P1/35]**

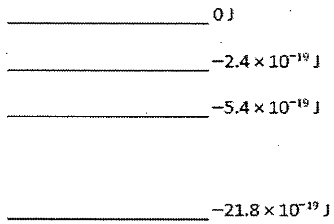
A beam of light of wavelength λ is incident normal to a surface of area S and is completely absorbed by the surface. The rate of photons arriving at the surface is n.

What is the pressure exerted on the surface by the light?

A $\dfrac{nh\lambda}{S}$ B $\dfrac{nh}{\lambda S}$ C $\dfrac{2nh\lambda}{S}$ D $\dfrac{2nh}{\lambda S}$

11. **[SAJC/H2/Prelims 2010/P1/35]**

The diagram shows the energy levels of a gas atom.



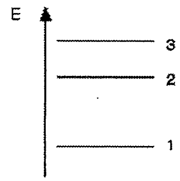
A free electron of kinetic energy of 20.0×10^{-19} J collides with the cool gas atoms.

What is the kinetic energy of the free electron after the collision?

A 1.8×10^{-19} J B 3.6×10^{-19} J C 5.4×10^{-19} J D 16.4×10^{-19} J

12. **[VJC/H2/Prelims 2010/P1/34]**

Part of the energy level diagram of a certain atom is shown. The energy spacing between levels 1 and 2 is twice that between 2 and 3. If an electron makes a transition from level 3 to level 2, the radiation of wavelength λ is emitted.



What possible radiation wavelengths might be produced by other transitions between the three energy levels?

A Only $\dfrac{\lambda}{2}$ B Both $\dfrac{\lambda}{2}$ and $\dfrac{\lambda}{3}$ C Only 2λ D Both 2λ and 3λ

13. **[YJC/H2/Prelims 2010/P1/31]**

Transitions between the lowest three energy levels in a particular atom give rise to three spectral lines of wavelength, in order of increasing magnitude, λ_1, λ_2 and λ_3.

Which of the following correctly relates λ_1, λ_2 and λ_3?

A $\lambda_1 + \lambda_2 = \lambda_3$ B $\lambda_1 = \lambda_2 + \lambda_3$ C $\dfrac{1}{\lambda_1} = \dfrac{1}{\lambda_2} + \dfrac{1}{\lambda_3}$ D $\dfrac{1}{\lambda_1} + \dfrac{1}{\lambda_2} = \dfrac{1}{\lambda_3}$

14. **[YJC/H2/Prelims 2010/P1/32]**

White light from a hot source is passed through sodium vapour and viewed through a diffraction grating.

Which statement best describes the spectrum seen?

A Dark lines on a white background

B Coloured lines on a white background

C Coloured lines on a dark background

D Dark lines on a coloured background

There are no options provided for the following questions.

15. **[YJC/H2/Prelims 2010/P1/34] (Amended)**

The de Broglie wavelength of an electron is 7.3×10^{-9} m with an uncertainty of 0.1×10^{-9} m.

What is the uncertainty in its position?

16. **[AJC/H2/Prelims 2010/P1/36] (Amended)**

The position of an electron is measured to the nearest 0.50×10^{-10} m.

Calculate its minimum uncertainty in momentum.

17. **[IJC/H2/Prelims 2010/P1/37] (Amended)**

A proton has a kinetic energy of 1.00 MeV and its momentum is measured with an uncertainty of 1.00%.

What is the minimum uncertainty in its position?

II. Structured Questions

1. **[IJC/H2/Prelims 2010/P2/6]**

a. i. Electromagnetic radiation has a wave nature as well as a particulate nature. This is known as the wave-particle duality. Describe a situation in which particles can be shown to have a wave nature. **[3]**

ii. Calculate the wavelength of a particle of mass 1.82×10^{-28} kg when travelling with a speed which equals to 10% of the speed of light. **[2]**

iii. Electromagnetic radiation can also be considered as a transverse wave and hence it also exhibits wave-like behaviour such as polarisation.

Explain why polarisation is a phenomenon of transverse waves but not longitudinal waves. **[1]**

b. Fig. 1.1 illustrates a phenomenon known as the Compton effect, whereby an incident X-ray photon is scattered by an electron at rest. The wavelength of the scattered photon λ' is found to be **longer** than the wavelength λ of the incident photon.

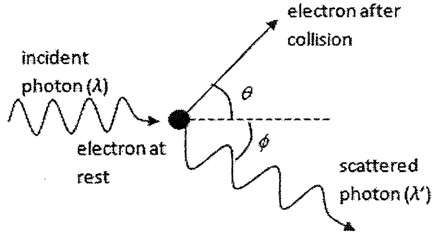


Fig. 1.1

Using de-Broglie's relation, suggest how this phenomenon demonstrates the particulate nature of electromagnetic radiation. **[2]**

[Total: 8]

2. **[IJC/H2/Prelims 2010/P3/7(part)]**

Fig. 2.1 represents the energy levels for an atom. The atom at ground state is bombarded with an electron of energy 17 eV.

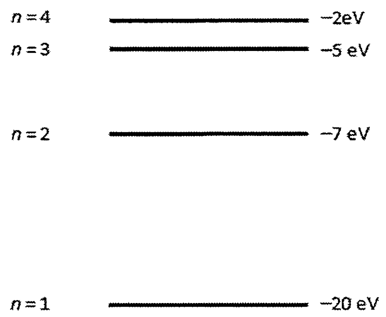


Fig. 2.1

a. State all possible photon energies when the atom returns to its ground state. **[2]**

b. On Fig. 2.2, sketch the appearance of the spectrum which corresponds to the frequencies of the emitted photons. **[1]**

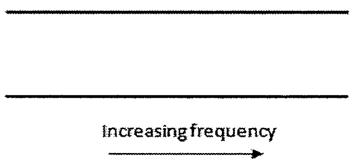


Fig. 2.2

b. State the difference between an emission line spectrum and an absorption line spectrum. **[2]**

[Total: 5]

3. **[MI/H2/Prelims 2010/P3/7]**

a. Explain clearly each of the following observations:

   i. Light waves seem to travel only in straight lines while sound waves and water waves can go around corners. **[2]**

   ii. Sound waves cannot be polarized but radio waves can. **[1]**

b. Fig. 3.1 shows some of the possible energy levels of an electron orbiting inside a mercury atom.

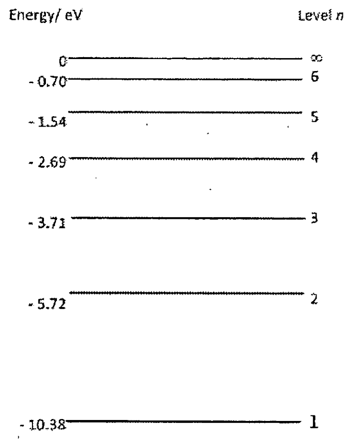


Fig. 3.1

   i. Explain how Fig. 3.1 can be used to account for the *emission line spectrum*. **[3]**

   ii. Cool mercury vapour is bombarded with a stream of electrons that have been accelerated from rest through a potential difference of 7.3 V.

      1. Determine the kinetic energy of an electron inside the stream. **[2]**

      2. Calculate the frequency of an emitted photon when an electron falls from Level 2 to Level 1. **[2]**

      3. State and explain whether photons with frequency in **b.ii.2.** would be emitted if electrons with energy of 4.15 eV collide with the mercury atoms. **[2]**

      4. On Fig. 3.1, draw the transitions that can be observed on the emission line spectrum when the mercury atoms are bombarded by electrons with energy of 7.3 eV. **[2]**

c. A particle of mass m and kinetic energy E has a de Broglie wavelength λ.

   Show that the expression for de Broglie wavelength λ in terms of m and E is given by

$$\lambda = \frac{h}{\sqrt{2mE}}.$$ **[2]**

d. An electron travels at 0.5 times the speed of light.

   i. Calculate its de Broglie wavelength. **[2]**

   ii. Comment and explain what is observed if such an electron beam is passed through a thin film of crystalline material. **[2]**

[Total: 20]

4. [NJC/H2/Prelims 2010/P3/3]

a. Describe how an emission line spectrum can be produced in the laboratory. Describe the appearance of the emission line spectrum when viewed through a grating spectrometer. **[3]**

b. i. The experiment below confirms that electrons occupied only discrete, quantized energy states.

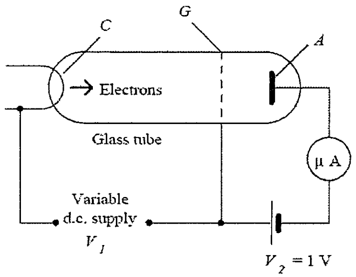


Fig. 4.1

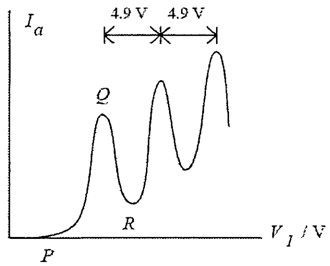


Fig. 4.2

Electrons emitted at the cathode C are accelerated by a potential difference of V_1 toward a positively charged grid G, in a glass tube filled with mercury vapor. Beyond the grid is an anode A, held at a voltage of V_2 of 1 V negative with respect to the grid (Fig. 4.1).

A graph of anode current I_a against V_1 is shown in Fig. 4.2.

The values of accelerating voltage where the current dropped gave a measure of the energy necessary to force an electron to an excited state.

Account for the shape of the graph when

1. V_1 is less than P. **[1]**
2. V_1 is between P and Q. **[1]**

ii. As shown in Fig. 4.2, when the accelerating voltage reaches 4.9 V, the current sharply drops, indicating the sharp onset of a new phenomenon. Suggest with explanation what the new phenomenon is. **[3]**

iii. 1. Using the values from the graph in Fig. 4.2, find the wavelength of the radiation emitted by the mercury atoms as they return to their ground state. **[1]**

2. State the region of the EM radiation which the wavelength calculated in **b.iii.1.** can be found. **[1]**

[Total: 10]

5. [RVHS/H2/Prelims 2010/P3/7(part)]

a. Fig. 5.1 shows a simplified representation of the 5 lowest energy levels of doubly ionised lithium (Li^{2+}) that has only one electron.

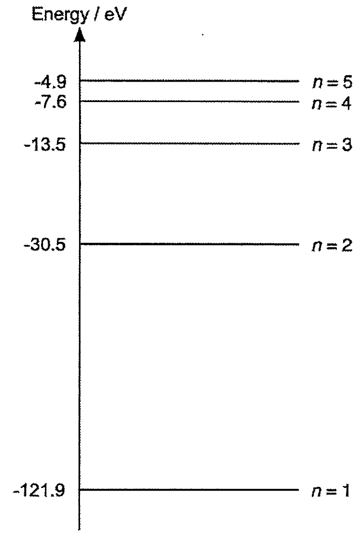


Fig. 5.1

i. Explain how emission spectral lines provide the evidence for the existence of discrete energy levels in an atom. **[3]**

ii. Explain why the ionised lithium vapour must be heated in order to produce an emission spectrum. **[1]**

b. Considering transitions between only these levels,

i. determine the wavelengths of the spectral transition that produces the shortest and longest wavelength **[3]**

ii. state the number of emission spectral lines that can be produced by transitions among these levels **[1]**

iii. sketch the emission spectrum of a gas consisting of these ions on Fig. 5.2. Use vertical lines to denote the relative positions of the spectral lines. **[4]**

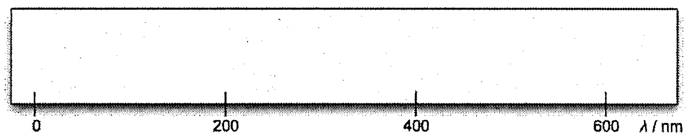


Fig. 5.2

c. Fig. 5.3 shows the electronic configuration of a lithium atom ($^{7}_{3}Li$).

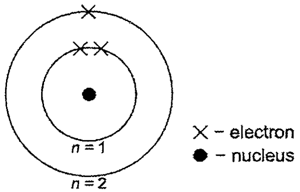


Fig. 5.3

The ionisation energies of a lithium atom are:

first ionisation energy: 5.42 eV

second ionisation energy: 76.0 eV

i. **1.** Explain what is meant by the term *ionisation energy*. [1]

2. State the value of the third ionisation energy. [1]

ii. The ionisation energy E of an atom can be used to estimate the radius of the atom. Using the uncertainty principle, estimate the radius of a lithium atom. [4]

[Total: 18]

6. [TJC/H2/Prelims 2010/P3/7(part)]

a. Fig. 6.1 shows a filament bulb emitting white light surrounded by a region of cooler helium gas.

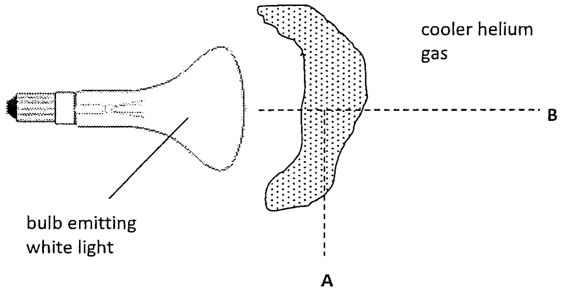


Fig. 6.1

State and explain the type of helium spectrum observed from

i. point A [2]

ii. point B. [2]

b. Fig. 6.2 shows some of the energy levels of an isolated atom of helium.

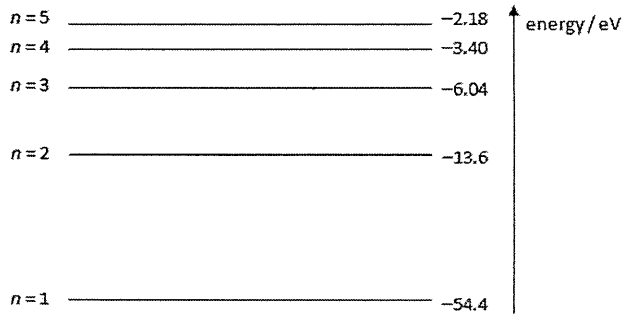


Fig. 6.2

i. An electron with kinetic energy 50 eV collides inelastically with a helium atom in the ground state. State the transitions which might take place following this collision. [2]

ii. Calculate the longest wavelength of the radiation that is emitted from these transitions. [2]

[Total: 8]

7. [VJC/H2/Prelims 2010/P3/7(part)]

a. Light of wavelength 540 nm is incident normally on a metal surface. The light has intensity 1.2 mW m^{-2} and the area of the metal surface is 1.4 cm^2. Calculate

i. the momentum of a photon of the incident light [1]

ii. the rate at which photons are incident on the metal surface [3]

iii. the force exerted by the light on the surface, assuming that all the light is absorbed [2]

iv. Suppose that the light is now incident on a piece of thin metal foil.
Explain whether the force calculated in **a.iii.** is a practicable means of moving the piece of foil. [2]

b. Although electrons, protons and neutrons are usually treated as particles, they also possess "wave" characteristics, which can be exploited by Transmission Microscopes to obtain high-resolution images of extremely small objects. For instance, electrons with a de Broglie wavelength of 5.0 nm can be used by such microscopes to image the structure of viruses.

i. Determine the kinetic energy of an electron which has a de Broglie wavelength of 5.0 nm. [2]

ii. The resolution of an image can be improved by using particles with shorter de Broglie wavelengths. Suggest two ways to decrease the de Broglie wavelength further, and explain your answer. [4]

[Total: 14]

Raffles Institution Physics Department Revision Package 1	Quantum Physics	19

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14
C	C	D	A	A	A	A	D	A	B	B	B	C	D

1 C With electrons of energy 12.9 eV, the hydrogen atom can only be excited to a maximum energy level of 0.85 eV. This is because –13.6 eV + 12.9 eV = –0.7 eV, hence it cannot be excited to –0.54 eV. Hence, by counting the number of possible emission spectral lines between the –0.85 eV, –1.51 eV, –3.39 eV and –13.6 eV energy level, there is a maximum number of 6 emission lines.

2 C Using de Broglie wavelength equation, $\lambda = \frac{h}{p} = \frac{h}{0.02(300)} = 1.1 \times 10^{34}\ m$

3 D When the cold gas is bombarded by electrons, the electrons in the gas atoms are excited to higher energy levels. The excited electrons remain there momentarily before de-exciting to lower energy levels. Upon de-excitation, photons are emitted with energy corresponding to the energy difference between the two energy levels. As the energy levels are discrete, the energy differences between the energy levels are also discrete. This emission of photons corresponds to the series of lines on the emission line spectrum.

4 A For monochromatic UV light, only one wavelength is produced. Hence, only possibility is that the atom is excited to n = 2 state, and de-excites from there with only one possible transition.

$$\Delta E = 13.6\left(1 - \frac{1}{2^2}\right) = 10.2\,eV$$

$$10.2\,eV = \frac{hc}{\lambda}$$

$$\lambda = 122\,nm$$

5 A From the transitions in the given diagram, there are two transitions that will emit high frequency photons of similar frequencies, and three transitions that will emit low frequency photos of similar frequencies. Hence, option A is correct as it shows two high frequency emission lines that are close to each other, and three low frequency emission lines that are close to each other.

6 A The free electron will lose energy as it excites the electron in the atom, exciting the atom's electron from its initial state to a higher state. Hence, the free electron will need an initial kinetic energy greater than the differences in energy levels of the atom.

7 A The difference in energy levels for the transition that produces λ_1 is smaller than that of the one for λ_2. Hence, the photon produced for λ_1 is will have a lower energy, and hence a longer wavelength.

8 D

$$P = \frac{NE}{t} = \frac{Nhc}{t\lambda}$$

$$\frac{N}{t} = \frac{P\lambda}{hc}$$

9 A From E_0 to E_1, the element is being excited. This is the result of the element absorbing energy, hence it should correspond to an absorption of radiation. Since the two energy changes in the question are the same, the element would be absorbing red light.

10 B Using de Broglie wavelength equation, $p = \frac{h}{\lambda}$

For N photons being absorbed in time t, the rate of change of momentum (and hence the force on the surface) is $F = \frac{Np}{t} = \frac{Nh}{t\lambda} = \frac{nh}{\lambda}$

Hence, pressure exerted is $P = \frac{F}{S} = \frac{nh}{\lambda S}$

11 B Energy level difference between the lowermost two levels

$$\Delta E = \left(-5.4 \times 10^{-19}\right) - \left(-21.8 \times 10^{-19}\right) = 16.4 \times 10^{-19}\ \text{J}$$

$$20.0 \times 10^{-19} - 16.4 \times 10^{-19} = 3.6 \times 10^{-19}\ \text{J}$$

12 B Other possible transitions include 2 to 1 (energy change of 2E) and 3 to 1 (energy change of 3E). These two transitions, with a higher energy change, will produce radiation of shorter wavelengths at $\lambda/2$ and $\lambda/3$.

13 C λ_1 corresponds to the transition from 3 to 1 (shortest wavelength, largest energy change). λ_2 corresponds to the transition from 2 to 1, and λ_3 corresponds to the transition from 3 to 2.

Therefore,

$$E_{31} = E_{21} + E_{32} \quad \Rightarrow \quad \frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3} \quad \Rightarrow \quad \frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$$

14 D When white light is passed through a cold gas, the photons with energies that correspond to the energy level differences in the sodium atoms are absorbed. This results in dark absorption lines. The diffraction grating disperses the white light into its different wavelengths, hence creating a coloured background.

15 From the de Broglie wavelength equation,

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad \frac{\Delta\lambda}{\lambda} = \frac{\Delta p}{p}$$

$$\Delta p = \frac{\Delta\lambda}{\lambda} p = \frac{0.1}{7.3} \times \frac{h}{7.3\times10^{-9}} = 1.24\times10^{-27}\ \text{kg m s}^{-1}$$

Using Heisenberg's uncertainty principle, $\Delta x \Delta p \geq h$

For minimum uncertainty, $\Delta x = \dfrac{h}{\Delta p} = 5.33\times10^{-7}\ \text{m}$

16 Using Heisenberg's uncertainty principle,

$\Delta x \Delta p \geq h$

For minimum uncertainty,

$$\Delta p = \frac{h}{\Delta x} = 1.3\times10^{-23}\ \text{kg m s}^{-1}$$

17

$$KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$p = \sqrt{2m(KE)} = 2.32\times10^{-20}$$

$$\Delta p = 0.01\left(2.32\times10^{-20}\right) = 2.32\times10^{-22}$$

Using Heisenberg's uncertainty principle, $\Delta x \Delta p \geq h$

For minimum uncertainty, $\Delta x = \dfrac{h}{\Delta p} = 2.86\times10^{-12}\ \text{m}$

Structured Questions

1. [IJC/H2/Prelims 2010/P2/6]

a. i. A beam of electrons passes through the graphite 'diffraction grating'.

An interference pattern of circular concentric rings is seen on the screen.

The de-Broglie wavelength of the electron is of the same order as the lattice spacing of the graphite atoms.

ii.

$$\lambda = \frac{h}{mv} = \frac{6.63\times10^{-34}}{\left(1.82\times10^{-28}\right)\times\left(0.10\times3.00\times10^{8}\right)} = 1.21\times10^{-13}\ \text{m}$$

iii. For a transverse wave, there are many possible vibrations of particles as long as they are perpendicular to the direction of transmission of the wave, but the axis of vibration of particles in a longitudinal wave is always parallel to the direction of transmission of the wave.

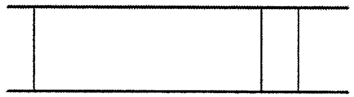
b. As a result of collision, the momentum of the electron increases and the momentum of the photon decreases as the momentum of the scattered photon $\frac{h}{\lambda'}$ is less than the momentum of the incident photon $\frac{h}{\lambda}$ (since λ' is longer than λ).

From the principle of conservation of linear momentum, the decrease in the momentum of the incident photon displays the particulate nature of electromagnetic radiation.

2. [IJC/H2/Prelims 2010/P3/7]

a. 2 eV, 13 eV and 15 eV.

b.



c. An emission line spectrum consists of bright lines against a dark background but an absorption line spectrum consists of dark lines against a continuous spectrum.

3. [MI/H2/Prelims 2010/P3/7]

a. i. Sound waves and water waves have wavelengths comparable to the size of the obstacles in their path. Sound and water waves and undergo diffraction around corners easily.

Light waves have wavelengths that are too small compared to the dimensions of the obstacles, and will not undergo diffraction. Thus light waves will appear to travel only in straight lines.

ii. Sound waves are longitudinal whereas radio waves are transverse. Only transverse waves can be polarized.

b. i. When the electron in a particular excited state falls to a lower energy state, it loses energy by emitting a photon.

Only photons of specific energies that are equal to the difference between two energy levels of the will be emitted.

Differences in energy levels $\Delta E = hf$, the frequencies corresponding to these photon energies give rise to an emission line spectrum.

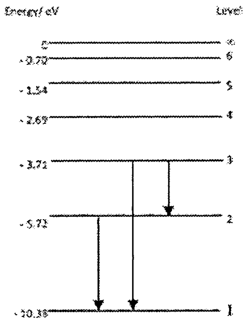
ii. 1. $W = eV \quad \Rightarrow \quad E_k = 1.60\times10^{-19}\times7.3 = 1.17\times10^{-18}$

2.

$$\Delta E = hf \quad \Rightarrow \quad f = \frac{\Delta E}{h} = \frac{\left[-5.72-(-10.38)\right]\times1.60\times10^{-19}}{6.63\times10^{-34}} = 1.13\times10^{15}\ \text{Hz}$$

3. Energy of electrons 4.15 eV is not sufficient to cause electron in the lowest level to move to next higher Energy Level 2 ($\Delta E = 4.66$ eV).

   Hence, it is not possible for photons of this frequency to be emitted.

4. 

c. $E = \dfrac{1}{2}mv^2 = \dfrac{m^2v^2}{2m} = \dfrac{p^2}{2m} \Rightarrow p = \sqrt{2mE}$

$p = \dfrac{h}{\lambda} \Rightarrow \dfrac{h}{\lambda} = \sqrt{2mE} \Rightarrow \lambda = \dfrac{h}{\sqrt{2mE}}$

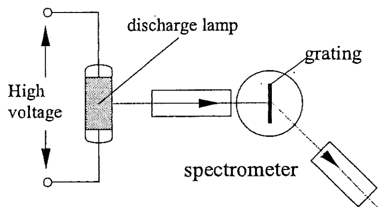
d. i. $\lambda = \dfrac{h}{p} = \dfrac{h}{mv} = \dfrac{6.63\times10^{-34}}{\left(9.11\times10^{-31}\right)\times0.5\times\left(3.00\times10^{8}\right)} = 4.85\times10^{-12}$ m

ii. The de Broglie wavelength of the electron is similar in size as the atomic spacing between atoms inside the crystal.

The spacing between atoms acts as a diffraction grating and diffraction effect of electrons will be observed.

4. **[NJC/H2/Prelims 2010/P3/3]**

a. A high voltage is applied across the discharge tube which contained the hot gases (for example, sodium vapour lamp or mercury vapour lamp). When the light emitted by hot gases in discharge tubes is passed through a diffraction grating or a prism, distinctive bright coloured lines against a dark background is observed. This is known as an emission line spectrum.



b. i. 1. When V_1 is less than P, the electrons reaching G do not have enough kinetic energy to reach A. Since no electrons reach A, I_a is zero.

Note: When electron moves from G to A, the electric force (repulsive force) will decelerate the speed of the electrons.

2. Electrons reaching G have enough kinetic energy to reach A. With increasing V_1, more electrons are able to reach A per unit time as they have higher kinetic energy, thus I_a increases.

ii. Kinetic energy from the accelerated electrons are passed to the mercury atoms (though inelastic collision) where electrons in the ground state are excited to a higher state. The sudden onset suggest that the mercury electrons cannot except energy until they reach the threshold for elevating them to an excited state.

iii. 1. $eV = \dfrac{hc}{\lambda} \Rightarrow \lambda = \dfrac{hc}{eV} = \dfrac{\left(6.63\times10^{-34}\right)\times\left(3.00\times10^{8}\right)}{\left(1.60\times10^{-19}\right)\times4.9} = 2.53\times10^{-7}$ m

2. Ultraviolet radiation

5. **[RVHS/H2/Prelims 2010/P3/7]**

a. i. Electromagnetic radiation is emitted as photons when electrons lose energy in the atom. The energy of the photons is hf, where f is the frequency of the radiation.

Spectral lines are discrete with well-defined frequency. Therefore electrons lose energy in discrete amount in atoms.

This is only possible if electrons in an atom transit between discrete energy levels.

ii. Heating causes the sizable number of ions to be excited, i.e., electrons are at higher energy levels.

When the electrons de-excite to lower energy states, photons with discrete frequencies are emitted.

b. i. $E = \dfrac{hc}{\lambda} \Rightarrow \lambda = \dfrac{hc}{E}$

Shortest wavelength: $\lambda_{min} = \dfrac{\left(6.63\times10^{-34}\right)\times\left(3.00\times10^{8}\right)}{\left(121.9-4.9\right)\times1.60\times10^{-19}} = 10.6$ nm

Longest wavelength: $\lambda_{max} = \dfrac{\left(6.63\times10^{-34}\right)\times\left(3.00\times10^{8}\right)}{\left(7.6-4.9\right)\times1.60\times10^{-19}} = 460$ nm

ii. Number of ways to produce spectral lines due to transitions of electrons between any two levels is ${}^5C_2 = 10$. Hence, there are <u>10 spectral lines</u>.

iii. Approx. correct position for spectral lines (5 → 1) and (5 → 4) at 10 nm and 460 nm

Approx. correct relative position for spectral lines (5 → 2) and (5 → 3)

3 other lines very close and to the right of (5 → 1) spectral line

2 other lines slightly further and increasing wavelength to the right of (5 → 2) spectral line

c. i. 1. The ionisation energy of an atom is the minimum energy required to remove an electron completely from the atom.

2. 122 eV

ii. 1st ionisation energy is 5.42 eV. Since

$$E = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE},$$

Lowest energy occurs when the momentum is the lowest but uncertainty principle imposes a limit to the least momentum. Consider the least momentum $\Delta p = p = \sqrt{2mE}$, using the uncertainty principle,

$$\Delta p \Delta x \geq h$$

$$\Delta x \geq \frac{h}{\sqrt{2mE}} = \frac{6.63\times10^{-34}}{\sqrt{2\times\left(9.11\times10^{-31}\right)\times\left(5.42\times1.60\times10^{-19}\right)}} = 5.27\times10^{-10} \text{ m}$$

6. [TJC/H2/Prelims 2010/P3/7(part)]

a. i. Point A – emission line spectra since the radiation is directly produced by excitation of He gas by photon absorption. Each radiation is a photon released when an electron falls from a higher energy level to a lower energy level.

ii. Point B – white light from the bulb contains continuous spectrum of photons of varying frequencies. Those photons corresponding to the energy difference between ground and excited states of He are absorbed and the transmitted beam having missing photons resulting in an absorption line spectrum.

b. i. The transitions for emission of radiation are: $n = 3 \rightarrow n = 1$, $n = 3 \rightarrow n = 2$, $n = 2 \rightarrow n = 1$.

ii. Longest wavelength came from $n = 3 \rightarrow n = 2$.

$$\Delta E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63\times10^{-34})\times(3.00\times10^{8})}{[-6.04-(-13.6)]\times1.60\times10^{-19}} = 1.64\times10^{-7}\text{ m}$$

7. [VJC/H2/Prelims 2010/P3/7(part)]

a. i.

$$p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{540\times10^{-9}} = 1.23\times10^{-27}\text{ kg m s}^{-1}$$

ii.

$$P = \text{intensity}\times\text{area} = (1.2\times10^{-3})\times(1.4\times10^{-4}) = 1.68\times10^{-7}\text{ W}$$

$$\frac{dn_{\text{photons}}}{dt} = \frac{P}{\frac{hc}{\lambda}} = \frac{1.68\times10^{-7}}{\frac{(6.63\times10^{-34})(3.00\times10^{8})}{540\times10^{-9}}} = 4.57\times10^{11}\text{ s}^{-1}$$

iii. Magnitude of the change in momentum of each photon as it is absorbed:

$$\Delta p = p_f - p_i = 0 - 1.227\times10^{-27} = 1.227\times10^{-27}\text{ kg m s}^{-1}$$

Force exerted on the surface: $F = \frac{dn_{\text{photons}}}{dt}\times\Delta p = (4.567\times10^{11})\times(1.227\times10^{-27}) = 5.60\times10^{-16}\text{ N}$

iv. Suppose the piece of foil has a mass of 1 g, the acceleration resulting from F would be

$$a = \frac{F}{m} = \frac{5.60\times10^{-16}}{1\times10^{-3}} = 5.60\times10^{-13}\text{ m s}^{-2}$$

This is extremely small, so the effect on the foil would be negligible. Hence, the force would not be a practicable means to move the piece of foil.

b. i.

$$E_k = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} = \frac{(6.63\times10^{-34})^2}{2\times(9.11\times10^{-31})\times(5.0\times10^{-9})^2} = 9.64\times10^{-21}\text{ J}$$

ii. The de Broglie wavelength of a particle may be expressed as $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE_k}}$.

If we increase the kinetic energy E_k of the particle, the above relation tells us that λ would be reduced.

So, the electrons can be accelerated through a larger potential difference to achieve a greater E_k, so that λ is smaller.

If we use a more massive particle (larger m), then λ would also be reduced.

So, instead of electrons, we can accelerate more massive particles like protons to achieve a shorter λ.